

Last Update: Aug. 20, 2026

EDUCATION

Georgia Institute of Technology

Ph.D. in Aerospace Engineering, Expected August 2027

Atlanta, GA, USA

Aug. 2024 – Aug. 2027 (Expected)

– Advisor: Prof. Koki Ho

Korea Advanced Institute of Science and Technology (KAIST)

Master of Science in Aerospace Engineering

Daejeon, Republic of Korea

Aug. 2022 – Aug. 2024

– Thesis: Optimal planetary surface routing considering synergy value system and heterogeneous agents

– Advisor: Prof. Jaemyung Ahn

Korea Advanced Institute of Science and Technology (KAIST)

Bachelor of Science in Aerospace Engineering

Daejeon, Republic of Korea

Mar. 2015 – Feb. 2019

Bachelor of Science in Physics (Double Major)

PROFESSIONAL AND RESEARCH EXPERIENCE

Georgia Institute of Technology

Graduate Research Assistant, Daniel Guggenheim School of Aerospace Engineering

Atlanta, GA, USA

Aug. 2024 – Present

Graduate Teaching Assistant (Spacecraft Flight Dynamics)

Aug. 2026 – Present

Korea Advanced Institute of Science and Technology

Graduate Research Assistant, Department of Aerospace Engineering

Daejeon, Republic of Korea

Aug. 2022 – Aug. 2024

Agency for Defense Development (ADD)

Researcher, Aerospace Technology Research Institute

Daejeon, Republic of Korea

Jun. 2019 - May. 2022

Research Officer for National Defense, Republic of Korea Air Force

PUBLICATIONS

Journal Articles

- [J1] **E. Choi** and J. Ahn, “Optimal Routing for In-Situ Resource Utilization-Based Planetary Surface Explorations,” *Journal of Aerospace Information Systems*, 2026 (Accepted)
- [J2] **E. Choi** and K. Ho, “Formulation and Analysis for Integrated Spacecraft Routing and Trajectory Design Problem,” *Journal of Spacecraft and Rockets*, 2026 (Accepted)
- [J3] B. Gwon*, **E. Choi***, J. Lee, and J. Ahn, “Adaptive Planning for Multiple UAVs with In-Flight Refueling,” *International Journal of Aeronautical & Space Sciences*, 2026, pp. 1–19
*These authors contributed equally.
- [J4] J. Choi, J. Lee, **E. Choi** and J. Ahn, “Sequential Vertiport Location Optimization Considering Uncertain Social Benefit,” *Journal of Aerospace Information Systems*, Vol. 23, No. 8, 2026, pp. 737–752
- [J5] **E. Choi** and K. Ho, “Orbital Depot Location Optimization for Satellite Constellation Servicing with Low-Thrust Transfers,” *Journal of Spacecraft and Rockets*, Vol. 63, No. 1, 2026, pp. 4–13.
- [J6] C. Moon, **E. Choi**, and J. Ahn, “Aerial-Ground Collaborative Routing with Uncertain Exploration Value,” *International Journal of Aeronautical & Space Sciences*, Vol. 26, 2025, pp. 2817–2841.
- [J7] J. Kim, **E. Choi**, J. Ahn, E.S. Suh, J. Kim, and D.G. Lim, “Mathematical Programming-Based Design Structure Matrix Clustering for Modular Architecture Design,” *Journal of Mechanical Design*, Vol. 147, No. 10, 2025, 101401.
- [J8] **E. Choi** and J. Ahn, “Optimal Multi-agent Grouping and Routing with Cooperative Tasks,” *Journal of Aerospace Information Systems*, Vol. 22, No. 3, 2025, pp. 163–176.
- [J9] **E. Choi** and W. Chang, “Multi-Agent Task Allocation with Inter-Agent Distance Constraints,” *Journal of Aerospace Information Systems*, Vol. 21, No. 2, 2024, pp. 168–177.
- [J10] **E. Choi** and J. Ahn, “Optimal Planetary Surface Exploration with Heterogeneous Agents,” *Journal of Spacecraft and Rockets*, Vol. 60, No. 4, 2023, pp. 1343–1354.
- [J11] **E. Choi** and J. Ahn, “Optimal Planetary Surface Routing Considering Exploration Values and Synergies,” *Journal of Aerospace Information Systems*, Vol. 19, No. 6, 2022, pp. 406–420.

Working Papers

- [P1] B. Gwon, **E. Choi**, and J. Ahn, “An Integer Linear Programming-Based Framework for Satellite Coverage Optimization Problem Under Operational Constraints,” *International Journal of Aeronautical & Space Sciences*, (Major revision)
- [P2] **E. Choi**, J. Lee, and J. Ahn, “Multi-Commodity Lunar Campaign Optimization via Leveraging Hybrid Propulsion Systems and Resonance Orbits,” *Journal of Spacecraft and Rockets*, (Major revision)
- [P3] **E. Choi**, J. Kim, J. Kim, J. Ahn, “Optimal Design Structure Matrix Clustering with Constraint Selections for Conflict Resolution,” *Journal of Mechanical Design*, (Under review)

Conference Papers

- [C1] **E. Choi** and K. Ho, “In-Space Recycling Logistics Optimization for Circular Space Economy,” *2027 AIAA SciTech Forum*, (Submitted).
- [C2] T. Noro, **E. Choi**, S. Han, M. Isaji, and K. Ho, “Optimal Scheduling of ADR Missions in the Near-GEO Region Based on a Competing-Risk Failure Model,” The 70th Space Sciences and Technology Conference, November 2026.
- [C3] T. Noro, K. Ho, **E. Choi**, S. Han, and M. Isaji, “Low-Thrust Trajectory Optimization via Sequential Convex Programming for On-Orbit Servicing in GEO,” The Japan Society for Aeronautical and Space Sciences 57th Annual Meeting, April 2026.
- [C4] **E. Choi** and K. Ho, “Optimal Routing and Trajectory Planning for Asteroid Mining with Partial In-Situ Resource Utilization,” *2026 AIAA SciTech Forum*, January 2026.
- [C5] **E. Choi** and J. Ahn, “Optimal Routing Problem with Precedence Constraints for Surface Exploration,” *2024 AIAA SciTech Forum*, January 2024.
- [C6] J. Choi, J. Lee, **E. Choi** and J. Ahn, “Optimizing UAM Vertiport Location in Jeju Island: Incorporating Noise for Public Utility,” *2023 Asia-Pacific International Symposium on Aerospace Technology*, October 2023.
- [C7] **E. Choi** and J. Ahn, “Optimal Short-Term Scheduling for Space Telescope in Low-Earth Orbit,” *2023 AAS/AIAA Astrodynamics Specialist Conference*, August 2023.
- [C8] **E. Choi**, D. Han, and W. Chang, “Decentralized Task Allocation for Multiple UAVs with Iterative Cost Reduction,” *KSAS 2023 Spring Conference*, pp. 60–61, April 2023.
- [C9] J. Ahn, **E. Choi** and D.U. Lee, “Application of routing problems to space exploration missions,” *2023 AIAA SciTech Forum*, January 2023.
- [C10] W. Chang and **E. Choi**, “Trade-off Study on Multi-UAV Autonomous Mission Planning Algorithms,” *KSAS 2022 Fall Conference*, pp. 1736–1737, 2022.
- [C11] **E. Choi** and W. Chang, “Multi-Agent Decentralized Task Allocation Algorithms with Fuel Constraints,” *KSAS 2021 Fall Conference*, pp. 432–433, 2021.

Media Articles

- [M1] **E. Choi** and K. Ho, “SpaceX aims to use rockets to quickly transport cargo across the Earth and into orbit – moves that would expand the global cargo system,” *The Conversation*, 2026.

PATENTS AND SOFTWARE DISCLOSURES

1. T. Noro, M. Isaji, **E. Choi**, S. Han, and K. Ho, “PosbasedOOSnearGSO.jl,” Software Disclosure, Invention ID 2026-248, 2025.
2. T. Noro, M. Isaji, **E. Choi**, S. Han, and K. Ho, “LowthrustSCP.jl,” Software Disclosure, Invention ID 2026-247, 2025.
3. D. Han, W. Jang, **E. Choi**, “Method and apparatus for determining the path of an unmanned vehicle using a distributed network,” *KR Patent*, 10-2946314, 2026.
4. **E. Choi**, K. Ho, and Y. Shimane, “Continuous Location Routing Problem for Orbital Depot (CLRPOD) v1.0.0,” Georgia Tech Software Disclosure, Invention ID 2026-018, 2025.
– <https://github.com/EuihyeonChoi/CLRPOD>

SCHOLARSHIPS AND AWARDS

- **Global Korea Scholarship Program for Study Abroad (Ph.D.) by Korean Government** Fall 2024 – Summer 2026
Annually \$40,000 and Airfare
- **Finalist in 2023 Space Grand Challenge by KAIST, Hanwha Aerospace, and Hanwha Systems** Nov. 2023
- **KAIST Scholarship** Fall 2022 – Spring 2024
Annually \$15,000 (Full-Ride)
- **ADD President’s Award: The Best Research Proposal** Dec. 2019
Subject: Battery Dumping System to Improve Drone’s Endurance
- **Degree of Bachelor of Science in Cum Laude** Feb. 2019
- **Finalist in NASA’s RASC-AL Competition** Cocoa Beach, FL, USA
Theme: Design Lunar Polar Sample Return Architecture Aug. 2017 – Jun. 2018
- **National Natural Science and Engineering Scholarship** Spring 2017 – Fall 2018
Total \$20,000 (Full-Ride)
- **The Boeing Company Scholarship** Spring 2016 – Fall 2017
Total \$3,500
- **National Scholarship for Undergraduate Study** Spring 2015 – Fall 2016
Total \$13,500 (Full-Tuition)

TECHNICAL JOURNAL OR CONFERENCE REFEREE ACTIVITIES

- Reviewer, *AIAA Journal of Aerospace Information Systems*, 2025, 2026
- Reviewer, *International Journal of Aeronautical and Space Sciences*, 2026
- Reviewer, *Cluster Computing*, 2026
- Reviewer, *AIAA SciTech*, 2025, 2026
- Reviewer, *AIAA Journal of Guidance, Control, and Dynamics*, 2025
- Reviewer, *The Journal of Supercomputing*, 2025

Mentoring & Professional Service

• Research Mentor (External) Mentoring a Master’s student on active debris maneuver and on-orbit servicing research	University of Colorado Boulder Apr. 2026 – Present
• Undergraduate Research Mentor Guided the implementation of orbital routing problems and paper replication for space logistics	Georgia Tech Spring 2026 – Present
• Invited Speaker Delivered an invited talk for the 6th Research Officer for National Defense	ADD Sep. 2022
• Selected Mentor Mentoring for the 4th Research Officer for National Defense	ADD Oct. 2020 – Jan. 2021

Projects

• Innovative Decision-Making Frameworks for Next-Generation Space Systems	Mitsubishi Electric Corporation, Georgia Tech Oct. 2025 – Aug. 2027
• Orbital Staging of Assets (Phase II)	Air Force STTR (Subaward from ThinkOrbital), Georgia Tech Oct. 2025 – July 2026
• ATLAS: Advancing Technologies for Logistics Architectures in Space	AFRL/AFOSR, Georgia Tech Aug. 2024 – Sep. 2026
• Space Logistics Model for Temporary Deployment of Earth Imaging Satellites	ThinkOrbital, Georgia Tech Feb. 2025 – Apr. 2025
• Mission Planning for Swarm of Tiny Satellites	LIG Nex1, KAIST Oct. 2023 – Jul. 2024
• Tactical Swarm Unmanned Aerial Vehicle Mission Planning and Autonomous Mission Replanning Technology	LIG Nex1, KAIST Oct. 2022 – Jul. 2024
• Modular Architecture Engineering: Extension Considering Practicality	Hyundai, KAIST Sep. 2022 – Jul. 2024
• Routing Problems Considering Value System for Exploration Missions	KAIST Sep. 2022 – Jul. 2024
• Low Observable Wingman UAV System	ADD Nov. 2021 – May. 2022
• Real-Time Decentralized Task Allocation for Multiple and Heterogeneous Unmanned Aerial Systems	AFRL, ADD Aug. 2021 – May. 2022
• Autonomous Navigation and Mission Management Technology	ADD Dec. 2019 – Aug. 2020